

How Our Hardest PDEs Coarse-Grain Reality: Yang–Mills, Navier–Stokes, and the Singularities a Finite Grain Forbids

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Preamble — what this article does NOT claim

1. **No new results, and no Clay proofs.** Every claim below is a standing corpus result, cited to its paper. This article does **not** solve the Yang–Mills mass-gap problem and does **not** prove Navier–Stokes global smoothness. Both remain open. What it does is locate each problem on one map and say, at honest tiers, how far the substrate reaches and exactly where it stops.
2. **The lens is a reading, not a theorem.** As in the companion article, “kept-column $\rightarrow$ variables, discarded-column $\rightarrow$ symmetries” is an organizing principle the corpus results instantiate; it is not itself derived from the primitives.
3. **Tiers vary by entry and are marked.** For Yang–Mills, the F^2 action is *derived given disclosed inputs* (structural/analytic), while the mass gap is *mechanism-tier* only. For Navier–Stokes, the coarse-graining to the fluid equations is *form-forced* with an inherited viscosity, and the no-blowup result is *structural-positive*, resting on declared postulates. None of these is a constructive proof.
4. **Two of the Navier–Stokes sources are different kinds of document.** The ED derivation (what the substrate builds) and the Architectural-Distillation evaluation (a structural scorecard of the *standard* nonlinear PDEs, built on the Factor Skyline framework, read *against* ED) are distinct lenses. They are kept distinct where they appear.
5. **The singularity-family section (§4) mixes executed results with same-shape conjectures.** ED’s finite-grain obstruction is carried to a structural-positive verdict in only two places — Navier–Stokes and gravitational collapse — and both rest on declared postulates. The other four singularity PDEs (NLS focusing, Keller–Segel, mean-curvature flow, Ricci flow) are flagged as conjectures of the same *form*, not results. The one genuinely new claim in this article is the **synthesis** that names a single finite-grain triple (P04, V5, P11) behind both executed cases; it is offered at synthesis tier, not as a theorem.

1. What Event Density is, and why these two are the “harder” shadows

Event Density (ED) is an account of physics built on a **discrete, relational substrate**: a participation graph in which chains participate in channels. Each channel carries a **bandwidth** $b \geq 0$ and a phase, combined in the participation amplitude $P = \sqrt{b} \cdot e^{i\pi}$. There is one process primitive: **commitment** (P11), the irreversible act by which an uncommitted participation becomes a definite, recorded fact.

The arrow of time is not a boundary condition on top of reversible laws; it *is* this act, written into the law. Everything else — space, geometry, matter, fields, the quantum formalism — is developed as structure in how committed events relate, as conditional derivations from thirteen posited primitives, with the form/value split always named (*form-forced, value-inherited*).

This is the companion to *How Our Best Theories Coarse-Grain Reality*, which read general relativity, quantum mechanics, Maxwell, Newton, and thermodynamics as blurs of that one substrate. That article's ledger is the tool here too: **every coarse-graining keeps a two-column ledger — what it keeps becomes the theory's variables, what it discards becomes the theory's symmetries**. This article applies the same tool to the two theories that are each tied to a famous *open* problem: Yang–Mills (the mass gap) and Navier–Stokes (global existence and smoothness). They are the “harder” shadows not because the lens works differently on them, but because the discarded column, in each case, is exactly the piece a Clay problem asks you to prove survives.

The two sit on *different rungs* of the same ladder, and naming the rung is half the chapter:

- **Yang–Mills is a channel-structure blur, kin to the Schrödinger chapter.** The gauge field is not an average of events; it is the structure of the channels themselves. It ties into the rope through the same channel/participation layer that gives ED charge and the amplitude.
- **Navier–Stokes is an average-type blur, one rung below general relativity on the same fluid ladder.** It is the hydrodynamic coarse-graining of the substrate's *transport* — the “stuff moving” sibling of GR's “shape” shadow.

After the two chapters, §4 widens the same finite-grain reasoning into a third thing: the family of *singularity-formation* equations — Navier–Stokes blow-up, the black-hole curvature singularity, and four PDE cousins — which the substrate meets with one shared move. The corpus lives at github.com/allen-proxmire/ED-generative (DOI 10.5281/zenodo.20149496); its front door is the Report, *Event Density: A Unified Framework for Physics*.

2. Yang–Mills — the force that talks to itself (*action derived; gap mechanism-tier*)

What it is a coarse-graining of. Not an average over events, but the *structure of the channels themselves*. In ED the gauge field is **P05 channel-transport** on a bundle: the base is emergent space, the fiber at each locus is the space of N dynamically-indistinguishable rule-type channels (P07), and the connection is how polarity is re-routed between them as one moves. The gauge symmetry is the receipt for one specific discard: **within a same-rule-type family, no substrate observable can resolve the channel tokens apart**. Blur those labels away and the fiber symmetry is $U(N) = SU(N) \times U(1)$ — the gauge group, read off the discard column. And the *action* is the same substrate viewed as a cost: the coherence-deficit on the $U(N)$ plaquette holonomy $1 - (1/N) \text{Re Tr } U_{\square}$ is the Wilson action, coarse-graining to $-\frac{1}{4} \int \text{Tr}(F^2)$.

- **KEPT** → the total participation bandwidth $\Sigma |\psi|^2$ (the sole accessible invariant, preserved as a P05 isometry); the *non-commuting* holonomy structure $[A, A]$; the F^2 kinetic term; and the *mass-gap mechanism* (whose survival into the continuum is the open Clay core, not a settled kept item, see below).
- **DISCARDED** → the channel token identities (this discard is what makes it a gauge theory); the retarded, non-hermitian arrow (averaged away to leave the OS-positive hermitian theory); and, not manufactured at all, the *specific* empirical group $SU(3) \times SU(2) \times U(1)$ and the couplings, which are inherited.

The mass gap: the force that talks to itself. For light — the abelian case, $U(1)$ — the channels

commute, so F is purely quadratic, the field does not source itself, and it stays massless. For the strong force — non-abelian $SU(N \geq 2)$ — the channels **do not commute**, so F carries the $[A, A]$ commutator, and the field gains cubic and quartic self-interaction. *The gauge field sources itself.* In the corpus’s own words: “Maxwell is massless because its channels commute; pure Yang–Mills gaps because they do not.” ED then reads the gap as a **finite correlation-length floor**: a finite reach means $\Delta \sim 1/\ell_*$, and a finite correlation length is a mass gap — “the gap is one over the floor.” The mechanism is checked numerically: the flat direction costs exactly 0.0000 for $U(1)$ and is lifted for $U(2), U(3), U(4)$.

The line ED does not cross. This splits into two tiers, and being precise about the split is the honesty:

- The **F^2 action is derived given two disclosed inputs** (the per-chain cost form and the effective-action / matter-averaging prescription) at the structural/analytic tier — the plaquette coherence-deficit gives $-\frac{1}{2}\text{Tr } F^2$, numerically verified. The corpus is explicit this is *derived-conditional*, not a from-nothing derivation.
- The **mass gap is mechanism-tier only**. ED locates *why* the gap exists; it does **not** prove the gap survives the continuum limit — and that survival is the non-perturbative core of the Clay problem. In the corpus’s grading: “Mechanism: verified. Clay proof: not attempted.” Continuum survival rests on a declared postulate (P-Profile-Rescaling) and hinges on asymptotic freedom, which ED does not prove; the coercivity and quartic-sign inputs (P-Gap-Coercivity, P-Quartic-Sign) are undischarged postulates, and the numerical value of Δ is not claimed.

There is also a standing **wall**: the multiplicities $\{1, 2, 3\}$ — why three families, why these group ranks — are **not** selected. ED’s own route to derive them was tested and refuted (a symmetric channel-multiplet is stable for *all* N), so this is a genuine open limit, not an oversight. The chapter’s Take is therefore the same shape as the general-relativity chapter’s: ED dissolves the *why* (the gap is non-commuting channels talking to themselves) and derives the action, but it does not hand you the continuum *proof*.

3. Navier–Stokes — the fluid is a shadow, and the grain leans on the singularity (*form-forced; smoothness structural-positive*)

What it is a coarse-graining of. The substrate’s *transport*, blurred in two steps. Substrate hops coarse-grain to ballistic worldlines carrying momentum — layer 1, the inertial/convective term $(\mathbf{u} \cdot \nabla)\mathbf{u}$, which ED builds natively. The further blur adds the viscous stress $\nu \nabla^2 \mathbf{u}$ — layer 2, the *same* velocity decorrelation that produced the diffusion recovery in the companion article, now carrying momentum, so viscosity ν is the momentum analog of the diffusion coefficient ($\nu \sim \ell^2/\tau$; scaling forced, prefactor inherited). Navier–Stokes is literally the “stuff moving” sibling of GR’s “shape” shadow, one rung down the same ladder, living in the hydrodynamic window $\ell_{\text{grain}} \ll R_{\text{cg}} \ll L_{\text{flow}}$.

- **KEPT** → the momentum-carrying convective term (layer-1 native); the Newtonian viscous stress (layer-2, measurable from the substrate, not merely inherited); and the finite-grain caps that bound how fast vorticity can grow.
- **ADDED at layer 2, the honest edge** → **incompressibility** $\nabla \cdot \mathbf{u} = 0$, an instantaneous, nonlocal elliptic constraint (the pressure-Poisson equation). ED is hyperbolic, generative, compressible; incompressibility is an admissible extension bolted on, “a limit, not a law,” not something the bare rule generates. External smooth forcing is likewise an input channel, not substrate-produced.
- **DISCARDED** → the discrete worldlines and commitments underneath. The substrate makes trajectories first; the smooth velocity field is a coarse-grained shadow of their density.

Does the grain forbid the singularity? This is the Clay question — can a smooth flow blow up in finite time — and here ED does something pointed: **the finite substrate grain structurally obstructs finite-time blow-up**. Three substrate caps compose. P04 (a bandwidth budget per cell) limits how fast vorticity amplitude can grow; V5 (a cross-chain budget) limits how fast a vortex line can stretch across cells; and P11 (commitment irreversibility) makes the obstruction monotonic — once a cell reaches its vorticity cap, the response is *dissipative*, not amplifying. Compose those with the standard Beale–Kato–Majda criterion and the vorticity integral stays finite: no finite-time singularity.

The line ED does not cross. The obstruction is a **structural-positive** verdict — “*not a constructive proof of global smoothness*.” It rests on two **declared** postulates (P-R1-Sufficient-Strength, P-Obstruction-Sufficient) and bounds the growth *rate*, not an absolute ceiling. Two further honesty pins:

- **The certified simulator does not itself relax to a fluid.** ED’s certified substrate is thin and ballistic; it does *not* thicken into the over-damped, many-collisions-per-step regime that a lattice-Boltzmann fluid needs. Navier–Stokes is reached in a *separate, postulated* hydrodynamic window, at the same structural/analytic tier as the gauge work — not on the certified simulator. “You reach the PDE only by leaving ED”: one substrate, two description-scales.
- **A note on the paper trail.** The long-horizon roadmap in the working repo (event-density/theory/Navier_Stokes_Roadmap_Scoping.md) was written 2026-04-28 as “*queued, not executing*” for the smoothness arc, predating the executed papers (Paper_084, Paper_077, Paper_099); it now carries a dated update noting NS-2 and NS-3 executed. The executed arc reaches structural-positive; the 2026-04-28 memo was the plan, not the result.

The second lens — where the whole problem localizes. Independently of the ED derivation, the Architectural-Distillation “PDE Atlas” scores the *standard* Navier–Stokes equation on six structural criteria, read against ED rather than derived from it. Its verdict earns a place here because it says where the difficulty actually lives. The scorecard splits cleanly 2D versus 3D: **Determinism passes in 2D and fails in 3D**, and that 3D failure, in the evaluation’s own words, “*is the Clay Millennium Problem*.” The composite reading is “*architecturally coherent, generatively powerful, but structurally non-optimal*,” with incompressibility flagged the same way ED flags it — “*non-minimal. This is a limit, not a law*.” The payoff line: the entire 2D/3D bifurcation “*reduces to a single sub-channel (A_stretch)... the architectural expression of the Millennium Problem*.” The whole famous problem localizes to **one channel: vortex stretching**.

That is exactly the channel ED’s grain-caps act on. The Atlas lens says “the whole problem is this one gate”; ED’s three caps are precisely a bound on *vortex-line stretching*. The two lenses meet on the same sub-channel from opposite directions — one localizing the open problem, the other supplying a substrate mechanism that leans on that exact gate, at structural-positive tier.

4. The singularity family — what a finite grain says about finite-time blow-up (*executed twice; the rest same-shape and open*)

The Navier–Stokes obstruction in §3 was not a one-off trick for fluids. It is one instance of a stance the substrate takes on *every* infinity: **a finite grain has no room to store one**. Nothing exists below ℓ_P to host infinite structural detail, so a true singularity — a point where some field runs to infinity — is categorically absent at the substrate level. Every famous finite-time-singularity problem then rephrases the same way: not “does the smooth equation blow up,” but “does the *continuum limit* re-introduce an infinity the substrate forbids?” The blow-up, if it is real, is a property of the blur, not of the world.

The finite-grain triple. The Navier–Stokes argument rested on three primitives: P04 (a per-cell bandwidth cap), V5 (a cross-chain cutoff, $\omega_c = c/\ell_P$), and P11 (monotone irreversibility, so a saturated cell dissipates rather than amplifies). That same triple is what forbids the *gravitational* singularity, and this is the load-bearing observation of the section: **the black hole has no singularity and the fluid does not blow up for the same structural reason, built from the same three primitives.** The roles are not identical in the detail. In the fluid, P04 caps a growth *rate* and V5 caps *spatial* vortex-line stretching, with vorticity the bounded quantity and the Beale–Kato–Majda criterion doing the closure; in the black hole, P04 caps a *cumulative total*, V5 is a *temporal* cutoff on the energy-concentration rate, and the bounded quantity is C_{cum} . So this is a shared finite-grain *template*, not one theorem run twice. The corpus built the two results independently and never wrote even the template down; here it is.

The gravitational member — and this one ED has worked out. Composing the triple, Paper_042_NoSingularity bounds the cumulative substrate commitment $C_{\text{cum}}(r)$ in a collapsing region: P04 bounds it, P11 makes it accumulate monotonically, and V5’s cutoff $\omega_c = c/\ell_P$ bounds the rate of energy concentration, so $C_{\text{cum}}(r)$ **stays finite as $r \rightarrow 0$** . In that paper’s words, “*the classical GR singularity is a coarse-graining artifact of treating substrate-level commitments as having no UV cutoff.*” The more detailed audit (Arc_BH_3) reaches the same verdict by gradient saturation: there is a maximum participation density ρ_{max} at which mobility vanishes — “*there is no $\rho \rightarrow \infty$ regime in ED*” — so “*singularities do not exist at substrate level; they are continuum-coarse-graining artifacts of the acoustic-metric approximation breaking down.*” What replaces the singular point is not a point at all but “*a finite-thickness saturated participation zone... not a curvature divergence,*” a region where the substrate gradients reach their maximum viable value and the smooth acoustic metric simply loses validity. Inside it every substrate quantity stays finite; only the continuum description fails.

Two honesty pins on the gravitational case. First, **the $b \rightarrow 0$ surface is the horizon, not the singularity** — a severance surface where capacity runs out, a different object from the saturated core. Second, the tier: the *replacement* is structural (forced at substrate level, with the values ρ_{max} and the critical gradient inherited), while Paper_042 rests on two declared postulates, and the corpus is explicit that only C_{cum} is shown bounded — *not every curvature invariant* — and that the interior structure below ℓ_P is open. The evaporation *endpoint* is a separate object again: a Planck-mass remnant (Paper_041), driven by the same ω_c cutoff but distinct from the interior result (and, the corpus warns, not the dark-matter candidate).

The rest of the family — same shape, not yet executed. The Architectural-Distillation atlas (built on Factor Skyline) catalogs the nonlinear PDEs and tags each one’s singularity type. Four of them are genuine singularity-formation equations: **NLS focusing** (blow-up above a mass threshold), **Keller–Segel** (the 8π chemotactic collapse), **mean-curvature flow** (a finite-time curvature singularity), and **Ricci flow** (whose neckpinch singularities, classified and surgically removed, are what settled the Poincaré conjecture). ED’s finite-grain stance predicts *no sub-grain singularity* in every case, with the continuum-re-introduction question left open — but the corpus has **not** carried out the obstruction for any of them, so they are flagged here as conjectures, not results.

The *kind* of prediction splits, though, and it matters. For Navier–Stokes and the black hole, the diverging quantity is one the substrate never builds, so the grain simply **forbids** it. For **Keller–Segel** and **Ricci flow**, the substrate *builds* the concentration — the corpus classifies chemotactic aggregation as layer-1-native structure-making (not a layer-2 forgetting like viscosity), and the neckpinch is the engine of Perelman’s proof — so the reading there is not “no singularity” but “the collapse **bottoms out at the grain** rather than reaching a true point”: a reframing, not a correction. Same finite grain, two verdicts, depending on whether the substrate builds the spike or merely fails to smooth it.

The corpus already draws a fluid $\leftrightarrow$ geometry link, but through a *different* ED obstruction than this section's, and the two must not be conflated. Paper_TheArrowSortsTheContinuum §7 notes that forming a *horizon* (the $b \rightarrow \emptyset$ surface) “has the same competition as Navier–Stokes: layer-1 concentration against layer-2 smoothing,” and calls the continuum-PDE map and the horizon map “two faces of one commitment primitive.” That link is Navier–Stokes $\leftrightarrow$ *horizon*, via the layer-competition. The one this section adds is a separate bridge: Navier–Stokes blow-up $\leftrightarrow$ interior *singularity*, via the finite-grain triple. Since the horizon and the singularity are different objects (the honesty pin above), these are two distinct bridges, not one endorsement of the other: the corpus supplies the first, this article the second.

Honest tier of the section: the general principle (no sub-grain singularity) is grounded in the finite grain; it is *executed* — carried to a structural-positive obstruction — in exactly two places, Navier–Stokes and gravitational collapse, both resting on declared postulates. The other four are same-shape and open. The unifying claim — one triple, many blow-ups — is a synthesis, new to this article, at synthesis tier.

5. The shared verdict

	What it is a blur of	KEPT	DISCARDED / the edge	The problem it meets	How far ED reaches
Yang–Mills	non-commuting channel (rule-type) transport; the plaquette coherence-deficit	bandwidth invariant; $[A, A]$; F^2 ; the gap	channel token labels ($\rightarrow$ gauge symmetry); the retarded arrow; the specific group $\{1, 2, 3\}$	the mass gap	F^2 action derived ; gap mechanism only; continuum survival open (asymptotic freedom unproven); $\{1, 2, 3\}$ a wall
Navier–Stokes	the substrate's momentum transport (kinetic $\rightarrow$ hydrodynamic)	convective $(u \cdot \nabla)u$; viscous stress; the finite-grain caps	the worldlines beneath; incompressibility is <i>added</i> , not native	global smoothness (finite-time blow-up)	fluid form forced , v inherited; no-blowup structural-positive (two declared postulates); not constructive

Both land at the **same tier** — what the corpus calls *structural-positive*: above a loose analogy, below a constructive proof. That is not a hedge; it is the identical tier as weak-field general relativity (the metric derived, the full field equations open) and the Born reconstruction (the pointer basis grounded, a rigor residual named) in the companion article. In every case ED does the same thing: it converts a “why is this even true” brute fact into a substrate *mechanism* — the gap is non-commuting channels sourcing themselves; the no-blowup is the finite grain refusing to be stretched past its cap — and then it stops honestly at the proof line.

What the lens *adds*, beyond the mechanism, is that it tells you **where the proof line is**. For Yang–Mills it is the continuum survival of the gap, which hinges on asymptotic freedom. For Navier–Stokes it is a single sub-channel, vortex stretching, which both lenses independently finger as the whole of the Millennium Problem. A substrate theory should not be expected to hand over a Clay proof; it should be expected to say precisely which discarded column the proof has to control. On these two, it does.

Honest close. Neither of these chapters is a prize claim, and the point of writing them is not the prize. It is that the same rope that ties GR and QM to one substrate reaches two more famous knots, at the same honesty as the two it already holds — and, in reaching them, points a specialist at exactly the sore spot in their own field: the continuum limit of the gap, and the vortex-stretching gate. And §4 shows the reach is wider than two knots: the finite-grain obstruction that leans on the fluid blow-up is built from the same three primitives that forbid the black-hole singularity, with four more singularity equations sitting in the same shape, unworked. The lens buys location, not victory. That is the honest thing it buys, and it is worth having.

Cross-references

- **Substrate & primitives:** physics-papers/foundations/Paper_087_13Primitives.md; the companion article Paper_HowCoarseGrainReality.md; the Report (repo root).
- **Yang–Mills — structure & action:** physics-papers/substrate-evaluation/Paper_GaugeStructure_FromChiralSU(N).md (SU(N), non-abelian, F^2 , hypercharge; $\{1, 2, 3\}$ wall); physics-papers/qft/Paper_018_YangMills1.md (the gauge field from multi-rule-type transport); working repo foundations/Gauge_06_YangMillsAction_FromChiralSU(N).md
- **Yang–Mills — the gap:** physics-papers/qft/Paper_021_YM4_MassGap.md (the correlation-length floor); working repo foundations/Gauge_07_YangMillsGap_FromCoherenceDeficit.md (gap $\iff$ non-commuting channels); physics-papers/qft/Paper_023_YM6_ClaySynthesis.md (Clay-relevance tiering); plain-language essays/ED_Essay_6_YangMillsMassGap.md.
- **Navier–Stokes — coarse-graining & viscosity:** physics-papers/soft-matter/Paper_076_NS2_CoarseGrainLayers.md (layers/layer_2/NavierStokes_LayerTwo.md (incompressibility as the edge); physics-papers/substrate-evaluation/Paper_Continuum_KineticLatticeGas.md (the certified substrate does not relax to a fluid).
- **Navier–Stokes — smoothness & blow-up:** physics-papers/soft-matter/Paper_084_VortexStretching.md (the three-cap obstruction); physics-papers/soft-matter/Paper_077_NS_Smoothness_R1.md (structural-positive verdict); physics-papers/foundations/Paper_099_NS_Synthesis.md (arc capstone & tiers); working repo theory/Navier_Stokes_Roadmap_Scoping.md (the 2026-04-28 scoping memo, since reconciled with a dated update noting NS-2/NS-3 executed).
- **The singularity family (§4):** the black-hole no-singularity result physics-papers/black-hole/Paper_042_NoSingularity.md (the P04+V5+P11 triple $\rightarrow C_{\text{cum}}(r)$ finite) and its detailed audit event-density/theory/Black_Holes/Arc_BH_3_Singularity_And_Strong_Curvature_Audit.md (the saturated participation zone, ρ_{max} floor); the UV cutoff physics-papers/black-hole/Paper_040_TransPlanckian.md ($\omega_c = c/\ell_P$); the evaporation endpoint physics-papers/black-hole/Paper_041_RemnantMass.md (Planck-mass remnant, distinct object); the corpus’s own fluid $\leftrightarrow$ horizon link in physics-papers/substrate-evaluation/Paper_TheArrowSortsTheContinuum.md §7 (“two faces of one commitment primitive”).
- **The Architectural-Distillation lens:** *An Architectural Atlas of Nonlinear PDEs* (A. Proxmire, 2026), the Factor-Skyline structural scorecard of the standard textbook PDEs (Architectural Distillation/evaluations/AD_Evaluation_PDE_Atlas/), read *against* ED rather than derived from it. It localizes the Navier–Stokes 2D/3D bifurcation to the vortex-stretching sub-

channel, and it is the source for the singularity-type tags on NLS, Keller–Segel, mean-curvature flow, and Ricci flow used in §4.